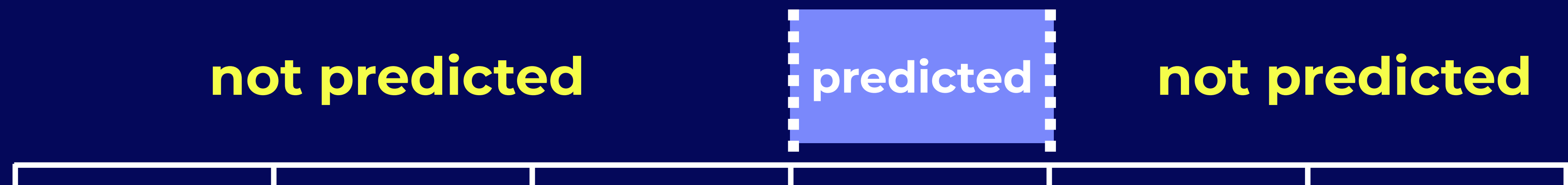
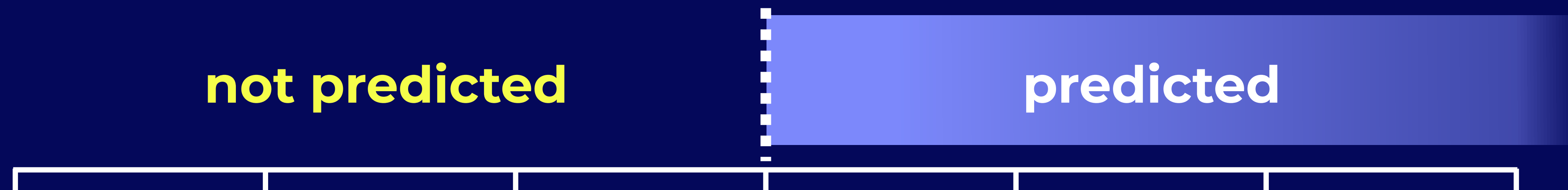
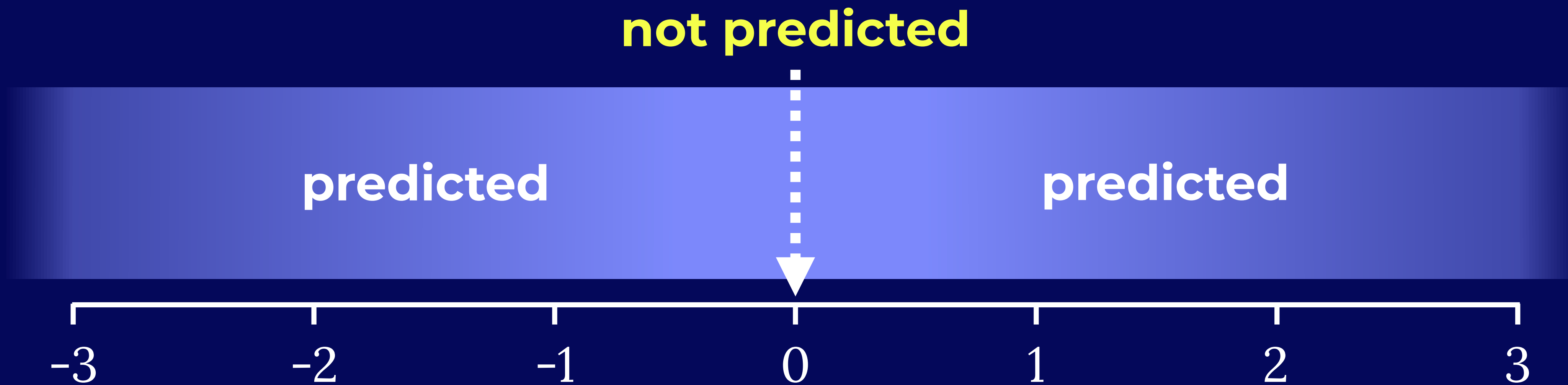


LING2136 / 4136

**Advanced Statistical Methods for
Language Students**

Session-10: SESOIs

Lecturer: Timo Roettger



not predicted

predicted

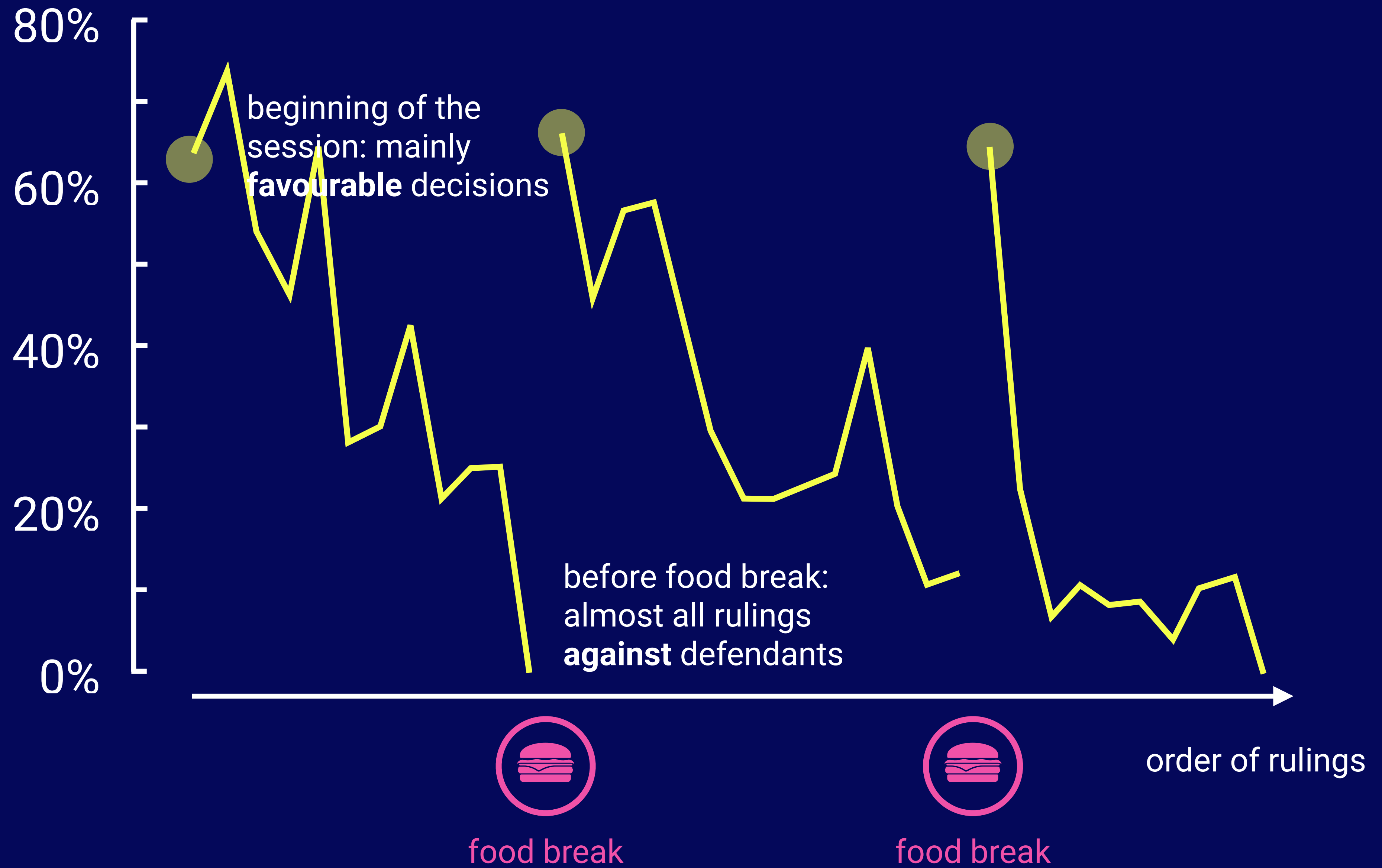
not predicted

Impossibly hungry judges?

Danziger et al. (2011), Lakens (2017)



% favorable decisions





emotional contagion?

Kramer et al. (2014)



There might even have been increased self-harm episodes, out of control anger, or dare I say it, **suicide attempts** or suicides that resulted from the experimental manipulation.

Nancy J. Smyth, 2014

reduced **positive**
emotional content

reduced **negative**
emotional content

reduced **positive**
emotional content

reduced **negative**
emotional content

reduced **positive**
emotional content

-0.1% positive words
+0.04% negative words

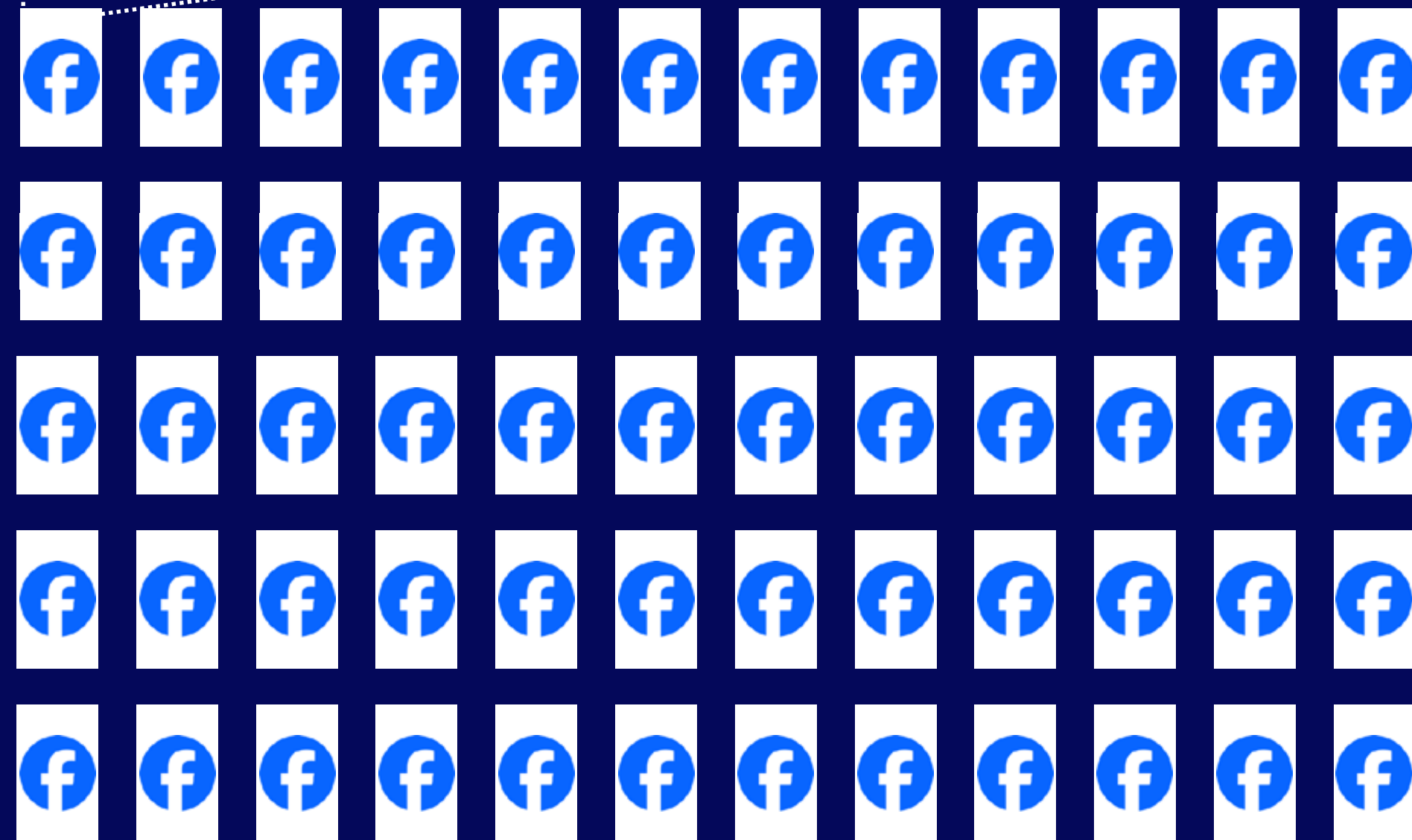
reduced **negative**
emotional content

+0.06% positive words
-0.07% negative words

reduced **positive**
emotional content

-0.1% positive words
+0.04% negative words

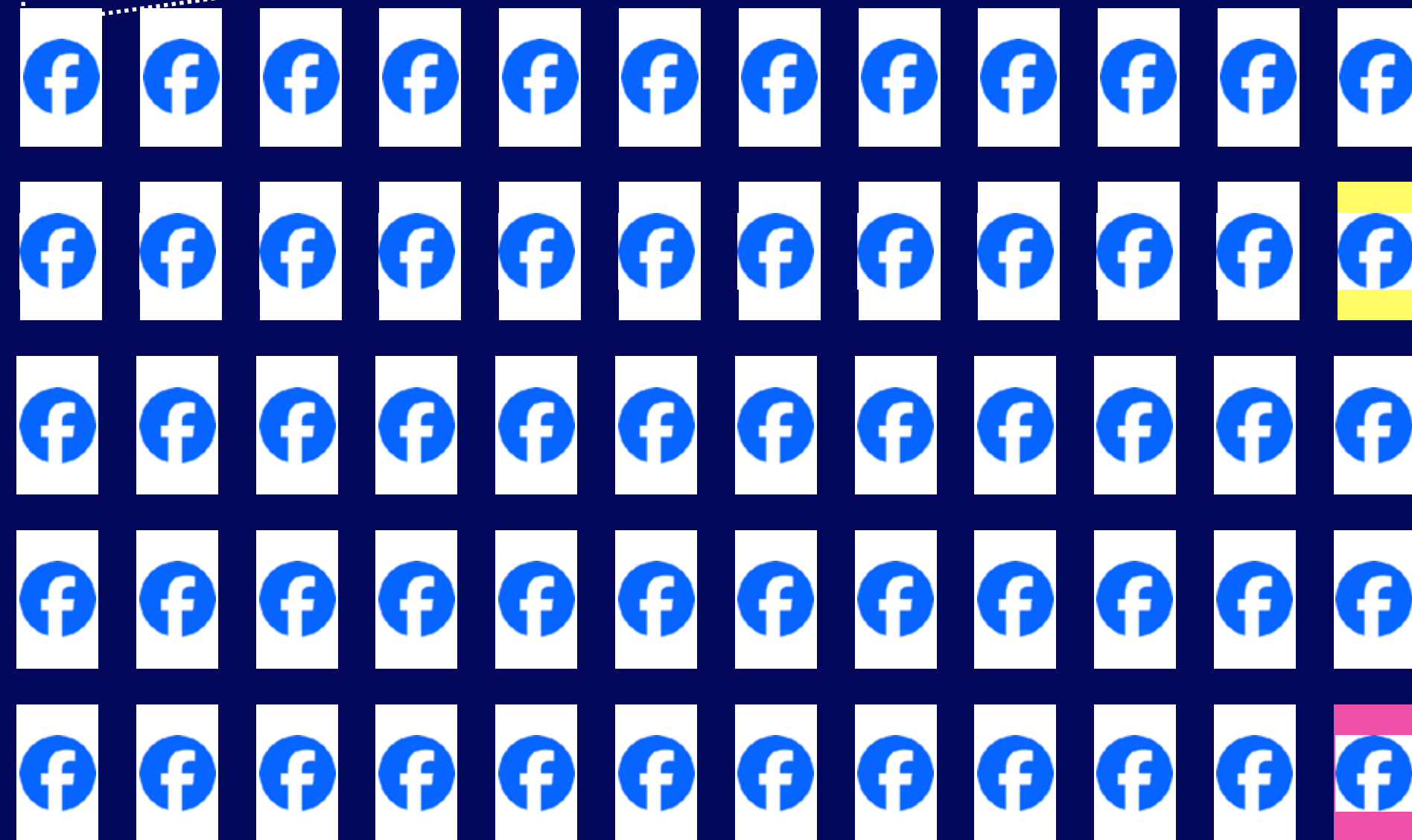
Today the pineapple clock decided to audition for my mailbox, and honestly the vibes were immaculate. I walked my toothbrush down the sidewalk while the clouds argued about spaghetti taxes. Everyone clapped, except the invisible hamster wearing sunglasses at the bar.



reduced **positive**
emotional content

-0.1% positive words
+0.04% negative words

Today the pineapple clock decided to audition for my mailbox, and honestly the vibes were immaculate. I walked my toothbrush down the sidewalk while the clouds argued about spaghetti taxes. Everyone clapped, except the invisible hamster wearing sunglasses at the bar.



-1 positive
word in 24
posts

+1 negative
word in 60
posts

reduced **positive**
emotional content

-0.1% positive words
+0.04% negative words

Today the pineapple clock decided to audition for my mailbox, and honestly the vibes were immaculate. I walked my toothbrush down the sidewalk while the clouds argued about spaghetti taxes. Everyone clapped, except the invisible hamster wearing sunglasses at the bar.



1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				



An average 20-year old in 2014
posted 2.4 posts a day Smith et al. (2017)



There might even have been increased self-harm episodes, out of control anger, or dare I say it, **suicide attempts** or suicides that resulted from the experimental manipulation.

Nancy J. Smyth, 2014

reduced **positive**
emotional content

-0.1% positive words
+0.04% negative words

in 2015: 18-24-year-olds
had (perhaps) around 650
facebook friends

2.4 posts per day x 650
friends = 1560 posts

Today the pineapple clock decided to audition for my mailbox, and honestly the vibes were immaculate. I walked my toothbrush down the sidewalk while the clouds argued about spaghetti taxes. Everyone clapped, except the invisible hamster wearing sunglasses at the bar.



62

- positive
words

156

+ negative
words

every day on your timelines



There might even have been increased self-harm episodes, out of control anger, or dare I say it, **suicide attempts** or suicides that resulted from the experimental manipulation.

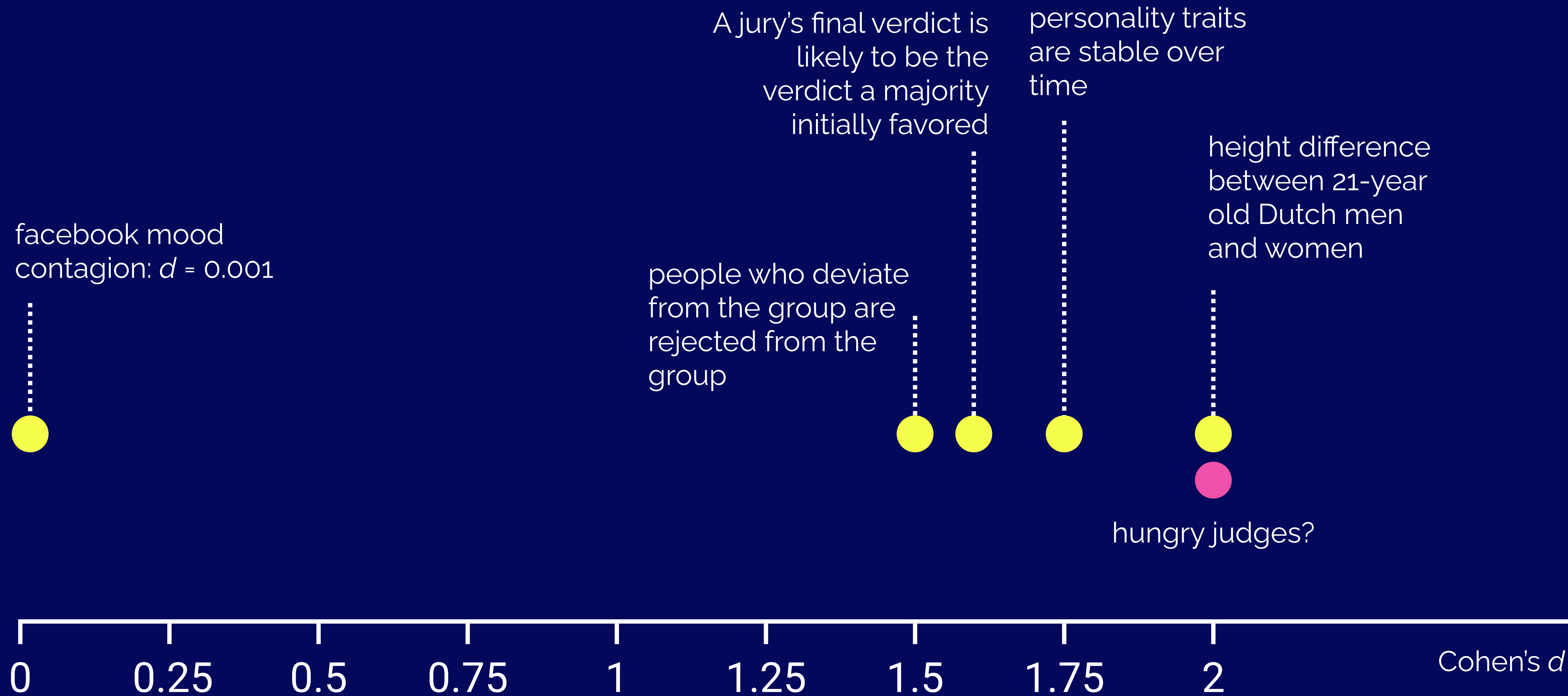
Nancy J. Smyth, 2014

We need to talk about **effect sizes**

$$d = \frac{\text{mean 1} - \text{mean 2}}{\text{standard dev.}}$$

simplified version

Cohen (1988)



facebook mood
contagion: $d = 0.001$



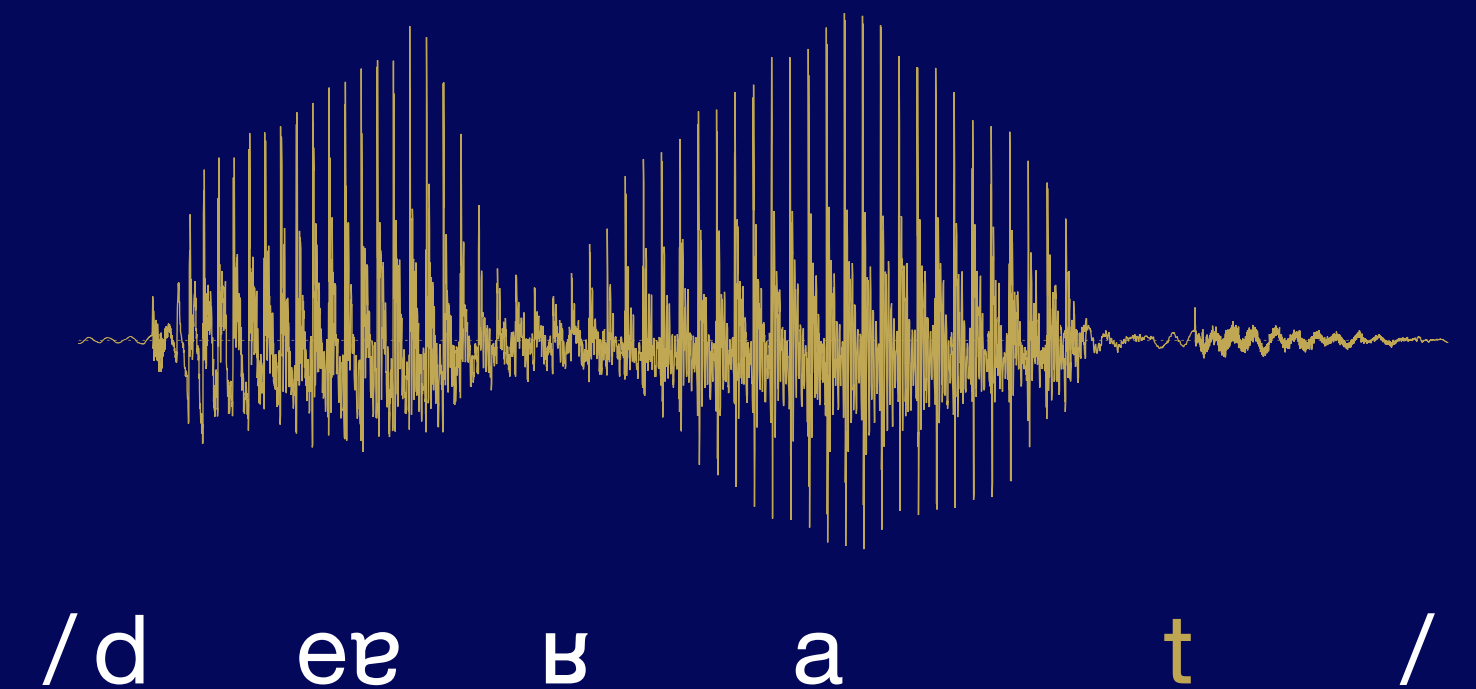
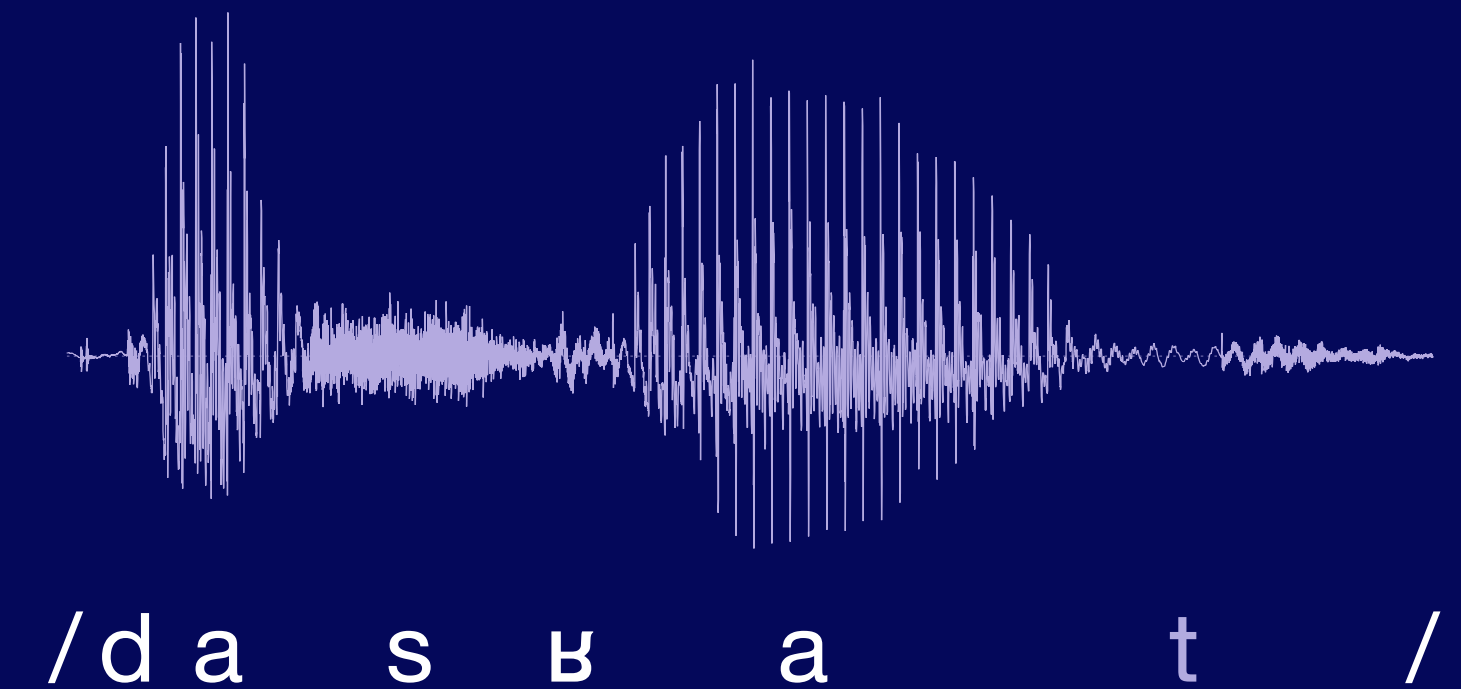
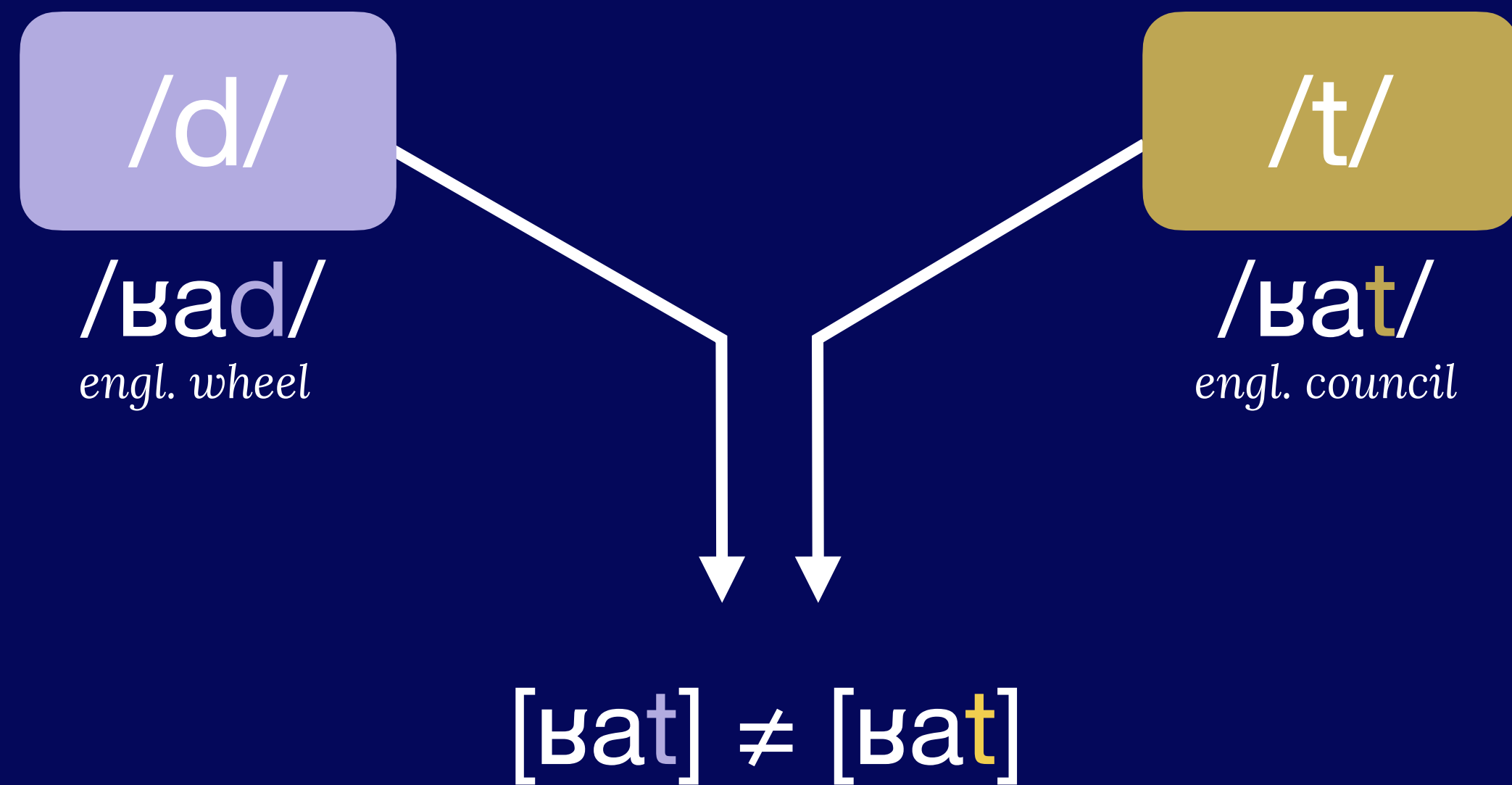
relevant effect magnitude?

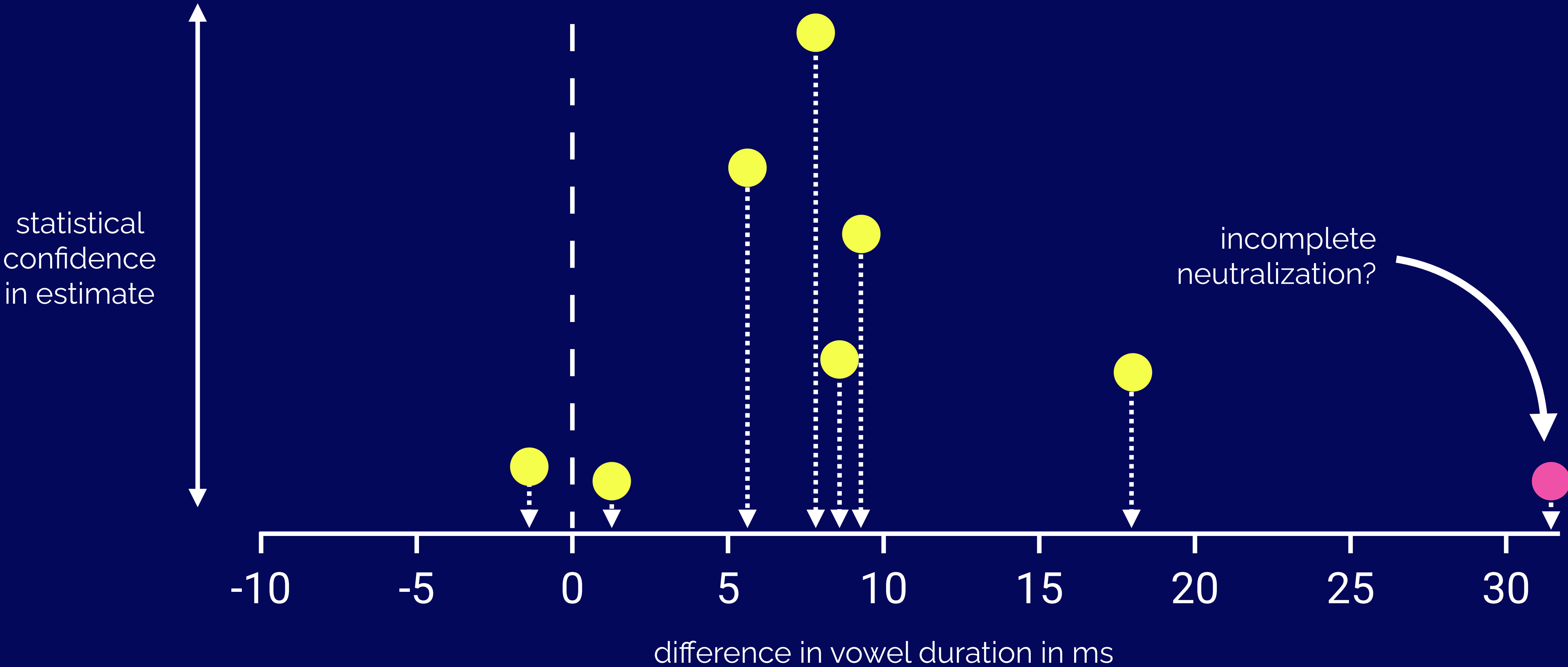
0 0.25 0.5 0.75 1 1.25 1.5 1.75 2

Cohen's d

incomplete neutralization

e.g. Port & O'Dell 1985, Fourakis & Iverson 1984, Kleber et al. 2010, Roettger et al. 2014, Niceboim et al. 2018

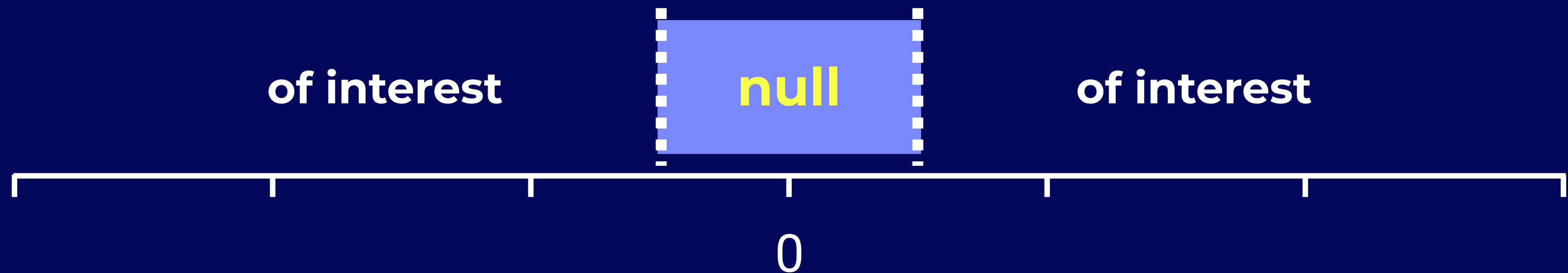




SESOI

Smallest Effect Size of Interest

Lakens 2017, Lakens et al., 2018



SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

Cost-Benefit analysis Peetz et al. (2025)

Minimal important difference Anvari & Lakens (2021)



SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

Cost-Benefit analysis Peetz et al. (2025)

Minimal important difference Anvari & Lakens (2021)

Just Noticeable Difference

“applied”

“theoretical”



SESOI

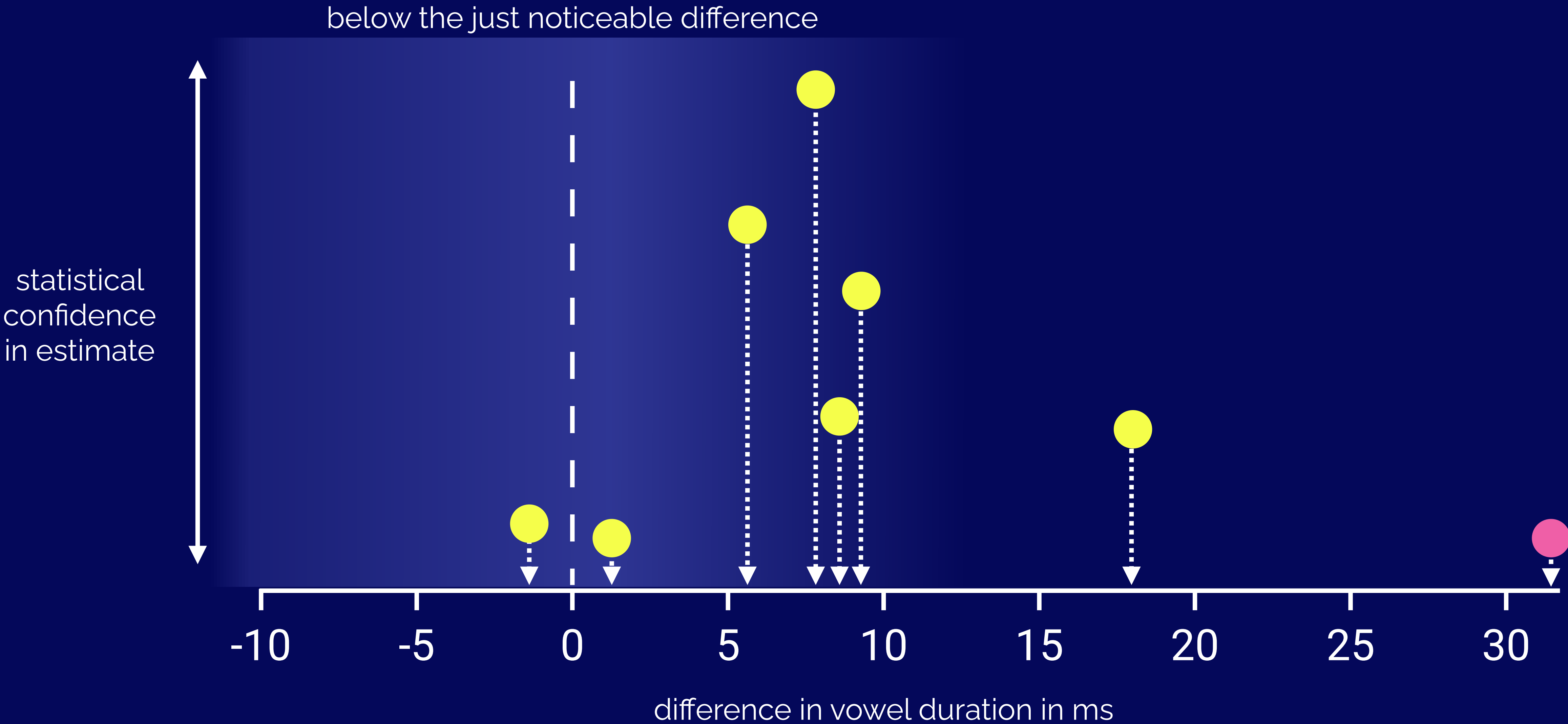
Just Noticeable Difference (JND)

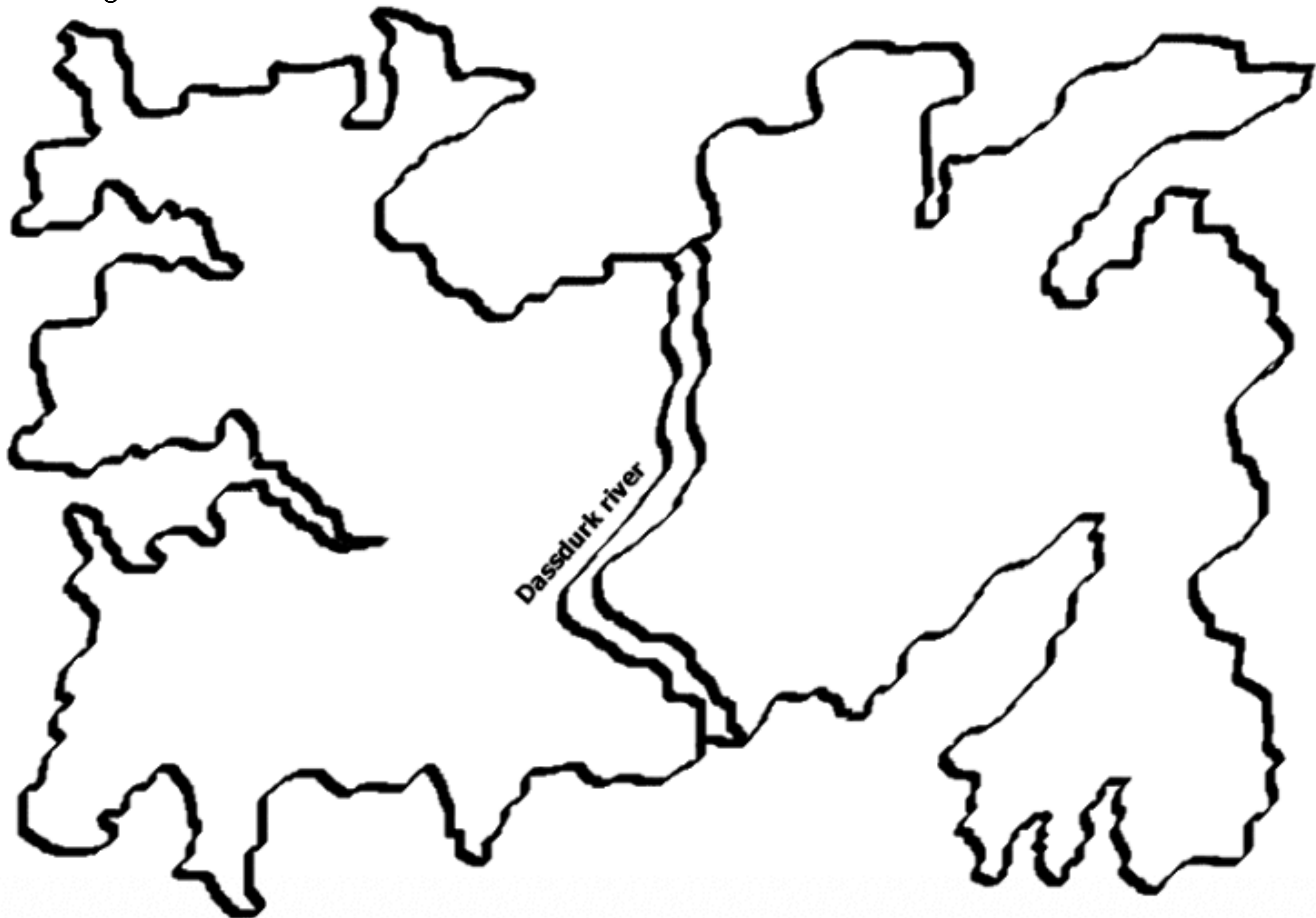
the minimum change in stimulus intensity (e.g., sound, weight, light) necessary for a person to detect a difference at least X% of the time (usually 75%)

Burriss et al. (2015)

increased facial redness in fertile phase -> attractiveness

effect was **too small** to be detectable by the visual system





SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

Cost-Benefit analysis Peetz et al. (2025)

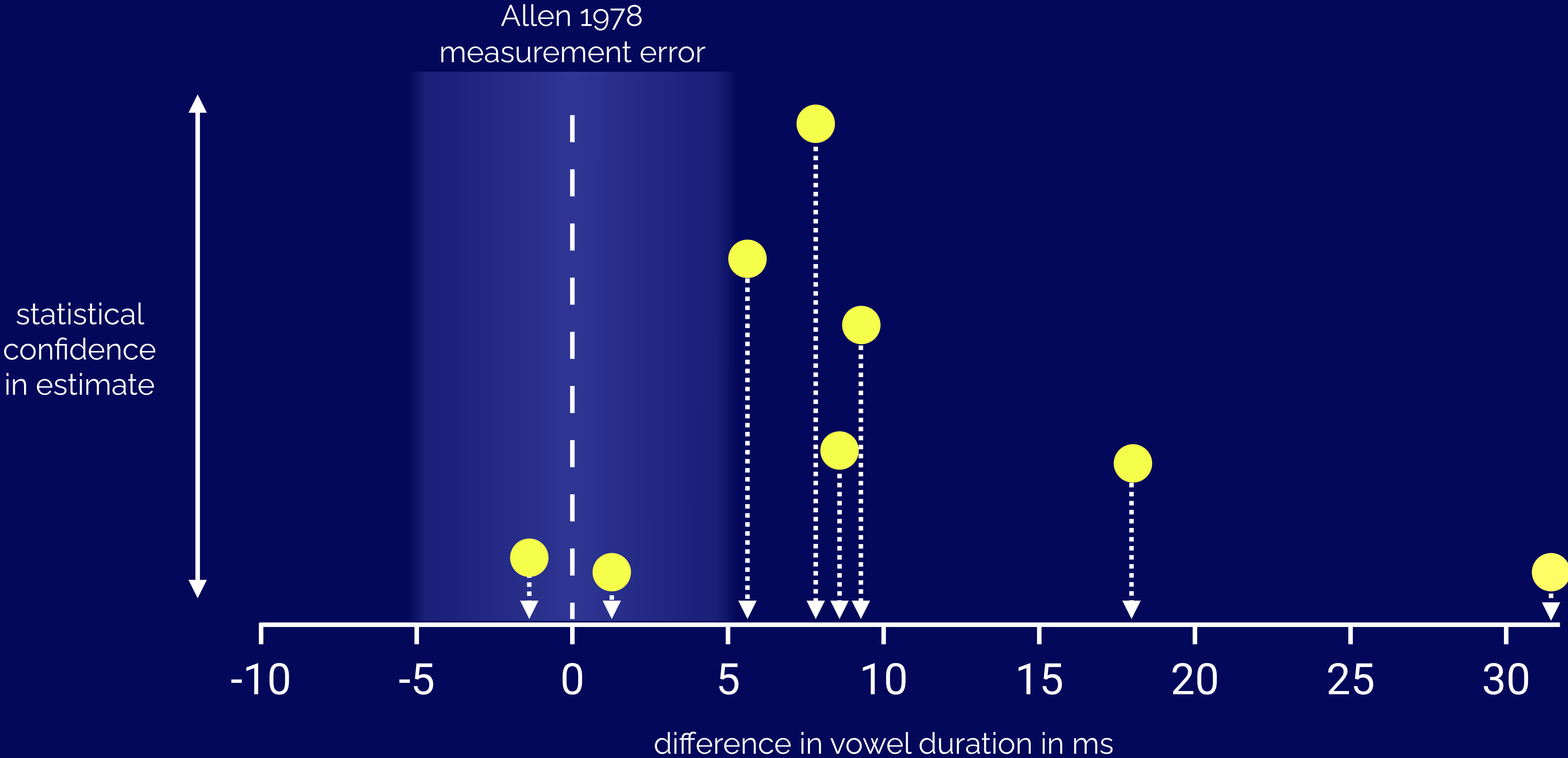
Minimal important difference Anvari & Lakens (2021)

Just Noticeable Difference

Measurement error

applied

“theoretical”

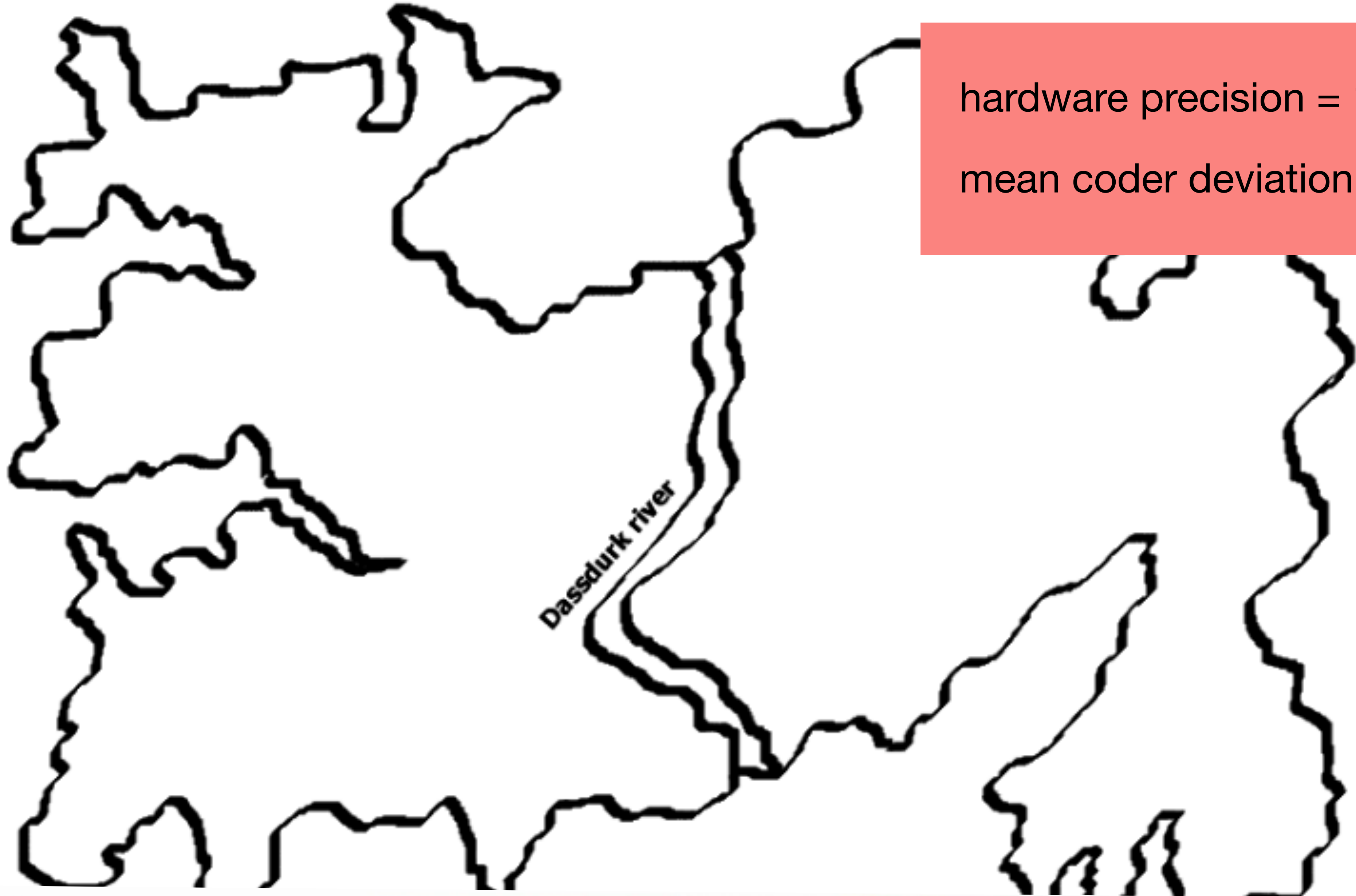


SESOI

Measurement error / limits

the minimum standard deviation for measuring vowel duration by phoneticians ranges from 2 - 5 ms (Allen 1978)





hardware precision = 1mm
mean coder deviation = 2.2mm

SESOI

Measurement error / limits

the minimum standard deviation for measuring vowel duration by phoneticians ranges from 2 - 5 ms (Allen 1978)

Winter & Matlock (2013)

A ruler has a precision of 1 mm.

mean deviation between people who measured: 2.2 mm



SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

Cost-Benefit analysis Peetz et al. (2025)

Minimal important difference

Anvari & Lakens (2021)

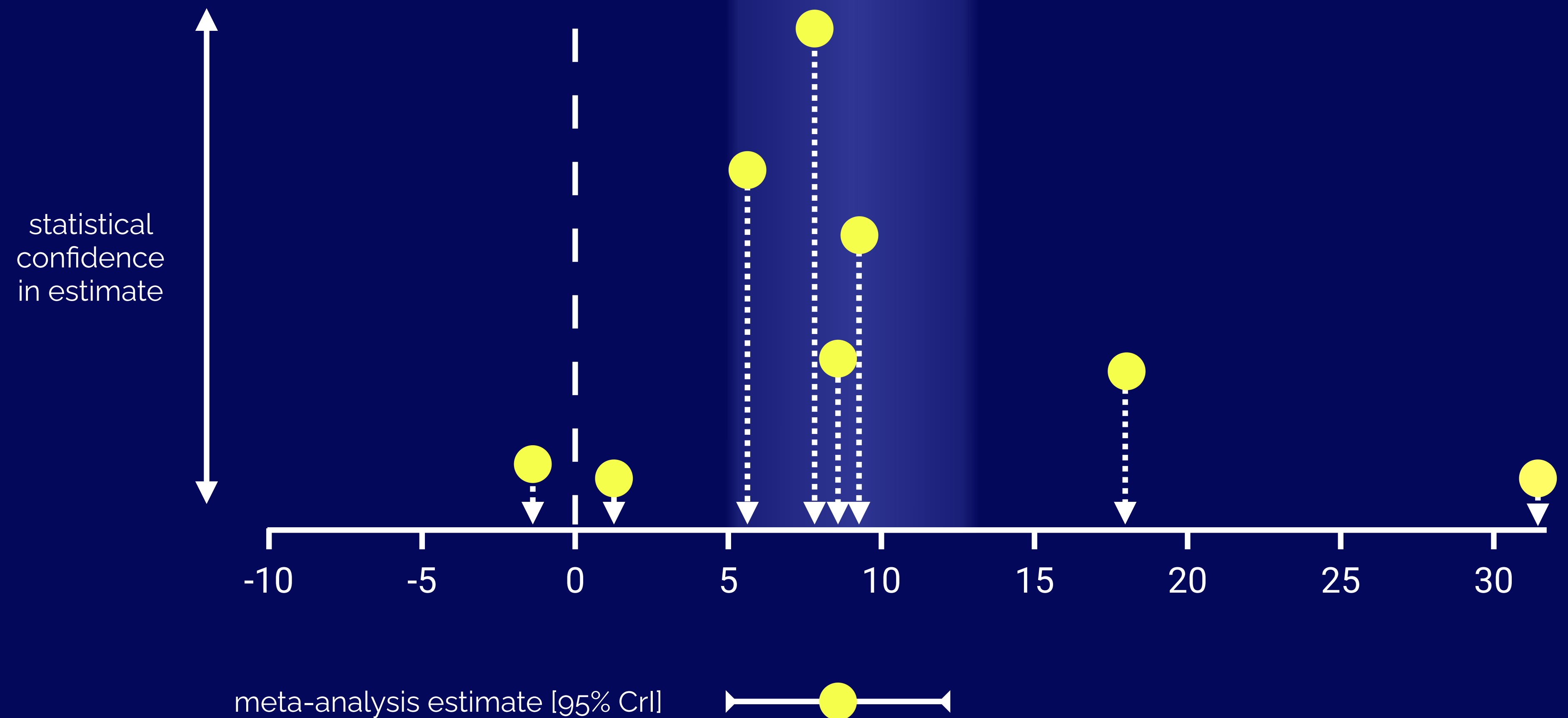
Just Noticeable Difference

Measurement error

Previous findings

SESOI

Based on
prior research



SESOI

Possible metrics

Person as effect size Peetz et al. (2025)

Cost-Benefit analysis Peetz et al. (2025)

Minimal important difference

Anvari & Lakens (2021)

Just Noticeable Difference

Measurement error

Previous findings

Expert opinions

SESOI

Based on expert opinion?



Timo B. Roettger @timoroettger.bsky.social · 5d

@lakens.bsky.social did you ever come across SESOI definitions for reaction time measures that are neither based on prior effect sizes or arbitrary standardized effect size thresholds? I am looking for a psychologically/methodologically motivated SESOI for RTs.

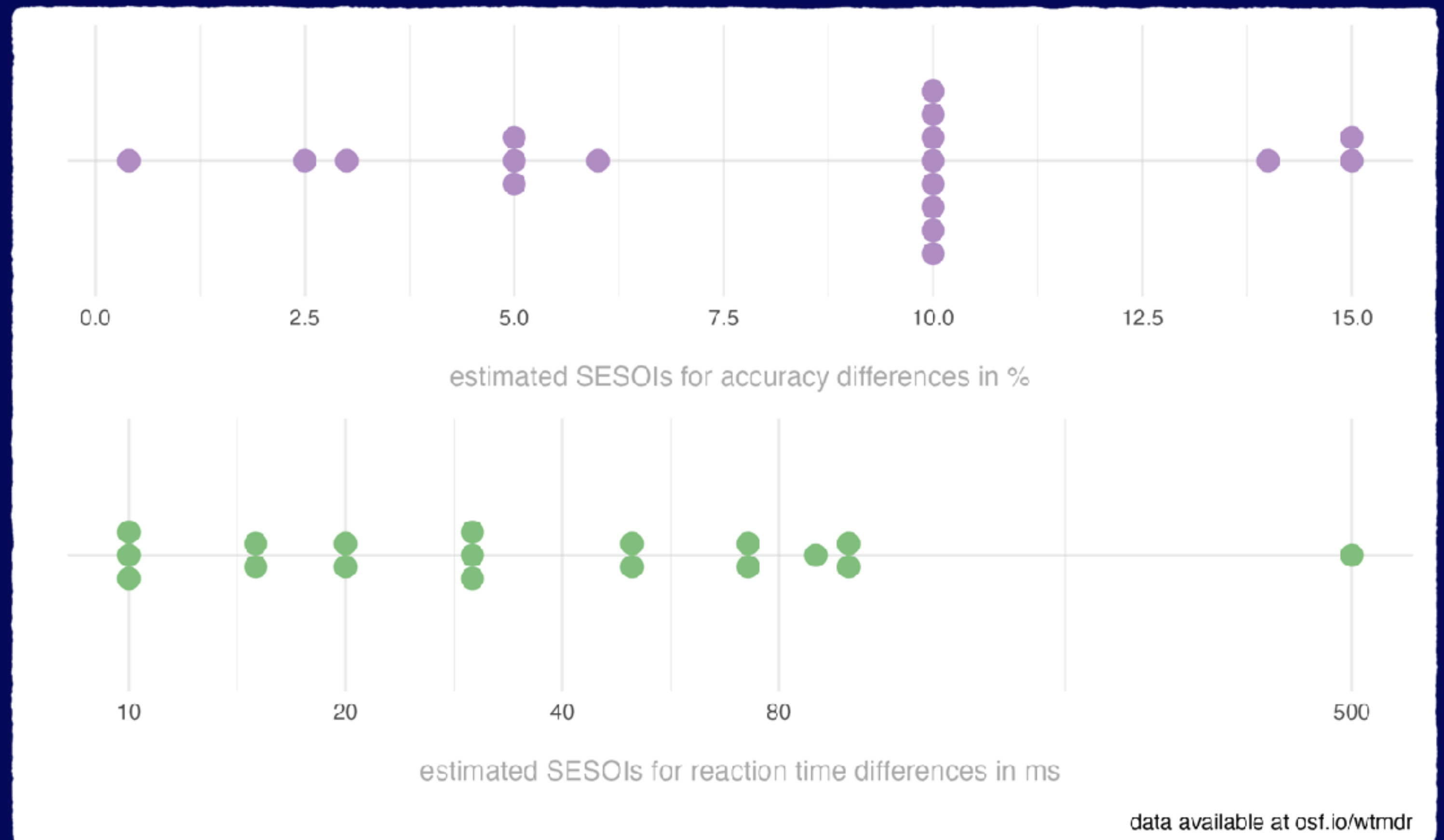


Daniel Lakens @lakens.bsky.social · 5d

We did it here, by asking original authors when the effect would be too small to theoretically be what they cared about
[pmc.ncbi.nlm.nih.gov/articles/PMC...](https://pubmed.ncbi.nlm.nih.gov/articles/PMC...)

SESOI

Based on expert opinion?



SESOL

honorable mentions

Imagine a researcher who wants to test if a reaction time game improves elderly adults' choice reaction time in a driving simulator. They determine that the smallest reaction time difference they would be interested in is the time it takes a car to travel 1 meter at a speed of 20 mph (32.1869 km/h), which is $1\text{ m} / \frac{32186.9\text{ m}}{3600\text{ s}} = 0.1118\text{ s}$. Roenker and colleagues (2003) report an average standard deviation of 0.268 seconds for choice reaction times in a driving simulator task in a sample of 55– to 86-year-olds (calculated as the mean of the nine standard deviations for Choice RTs in Table 1 in Roenker et al., 2003). Based on these data, 0.1118 seconds would correspond to a standardized mean difference of $d = \frac{0.1118\text{ s}}{0.268\text{ s}} = 0.42$.

SESOL

a decision tree?

Can you define a SESOL...

in an applied way? (e.g. cost-benefit analysis)

derived from theoretical principles? (e.g. JND)

based on measurement considerations? (e.g. precision)

based on previous findings? (e.g. meta analysis / expert opinion)

if you can't... try harder

or... you know... don't test hypotheses